



# Sensor-Based Airbag

Freejump - LEAPS Partnership Approach

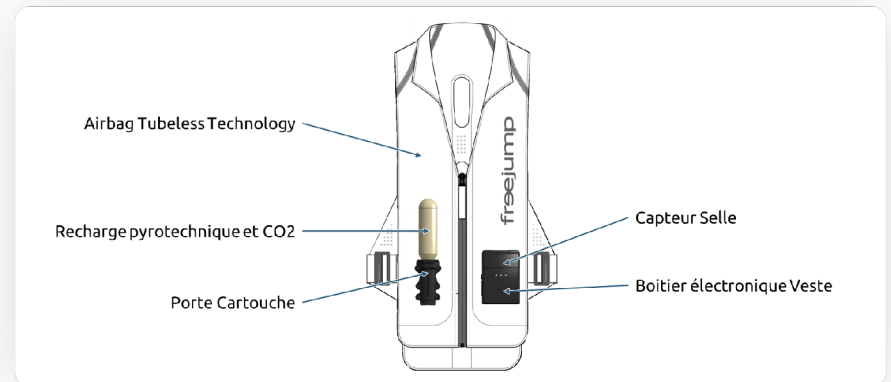
Freejump · April 2026

## SYSTEM

# Wireless Equestrian & Cycling Airbag

Build an airbag integrating **electronic fall detection** and a **pyrotechnic triggering system**. Fast, reliable, ergonomic protection with no mechanical tether to the saddle.

- Designed for riding, training, and competition
- **Real-time** fall detection before ground impact
- Inertial sensors + embedded algorithm
- Ultra-fast inflation via pyrotechnic gas generator



Freejump airbag vest — inner view

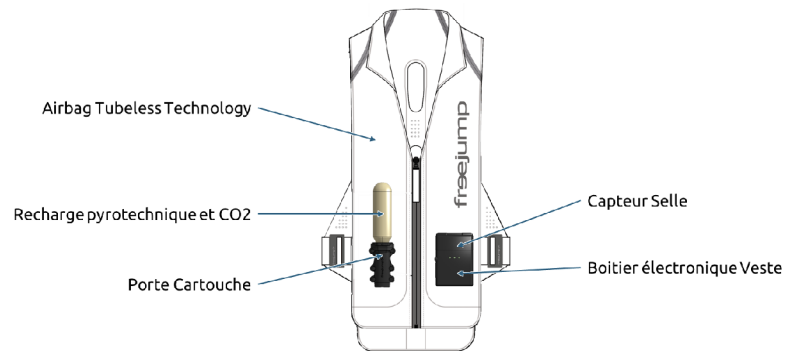
## PRODUCT

# 3 Key Components

1

## Tubeless Vest

Tubeless Technology



2

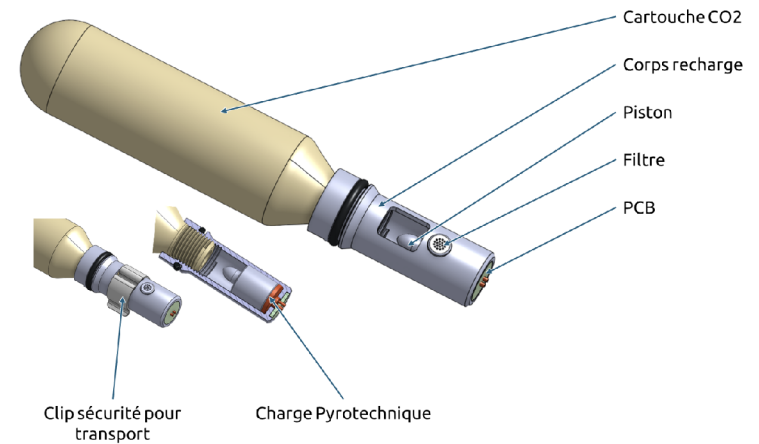
## Acquisition Unit

IMU + sensor fusion

3

## Gas Generator

Pyrotechnic cartridge



## OBJECTIVE

# Reliable Fall Detection in Real Time

- Complex environment: vibration, noise, and varied terrain
- Zero false positives to preserve user trust
- Embedded on inertial sensors + complementary technologies

### **Key constraint:**

A false positive means loss of trust.

A missed detection means real danger.

## CURRENT APPROACH

# Method We Are Using Today

- **Energy-based approaches** — gyro energy and acceleration as fall signatures
- **Statistical approaches** — sliding windows, adaptive thresholds, distributions
- **Combining both** — crossing energy and statistics improves reliability

We combine physics-driven indicators with statistical robustness to improve detection confidence in real riding conditions.

## TECH EXPLORATION

# UWB + IMU Fusion Opportunities

### UWB — distance measurement

- Real-time rider ↔ horse distance
- Centimeter-level precision
- Complementary signal to IMU-based detection

### Sensor fusion — 3D orientation

- Saddle and rider orientation estimation
- Automatic synchronization of both sensors
- Motion signature extraction for fall scenarios

Target architecture for the next phase:

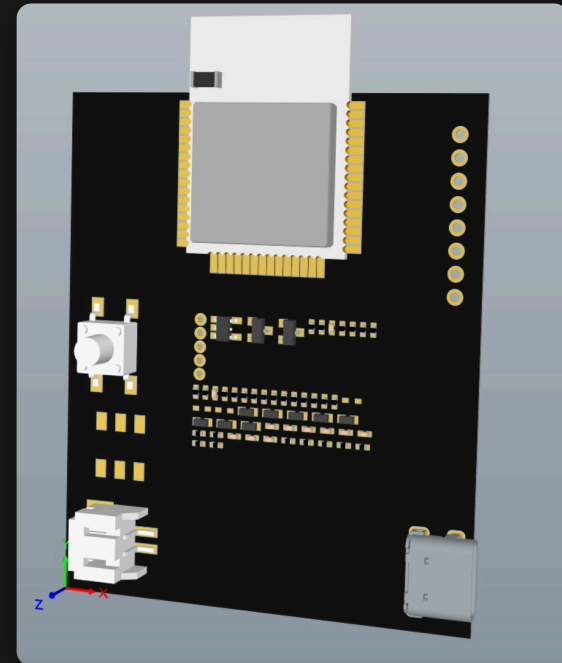
**IMU-only → IMU + UWB + orientation features.**

## PROTOTYPING

# Prototype Setup

- Based on **DWM3001CDK** evaluation board
- Add a **boost circuit** for pyrotechnic triggering
- Add an **IMU** aligned with our detection approach

Goal: quickly validate an integrated UWB + IMU platform ready for real-world fall-detection experiments.



WITH LEAPS

## What We Could Build Together

- Co-design a robust UWB + IMU hardware architecture for airbag use cases
- Define and validate reliable ranging + synchronization performance in riding conditions
- Accelerate industrialization path from prototype to production-ready module

We are looking for a strong technical partnership with LEAPS, leveraging Qorvo technology to secure performance and time-to-market.





# Thank You

Any questions?

## APPENDIX — PHASE 1



# Initial Approach — Alogo

- Analysis outsourced to a specialized external provider
- Threshold-based approach + combinatory rules on 2 sensors
- End-to-end pipeline: 8 steps from acquisition to trigger

Result: auditable logic and identified parameters,  
but too rigid for reliable production deployment.

## APPENDIX — KEY LEARNINGS

# What We Learned

### ✓ What Worked

- **Energy-based approaches** — gyro energy and acceleration as fall signatures
- **Statistical approaches** — sliding windows, adaptive thresholds, distributions
- **Combining both** — crossing energy and statistics improves reliability

### ✗ Why We Did Not Continue

- Threshold logic was too rigid for real-world variability
- No fast iteration capability
- Not production-ready in its current form

## APPENDIX — DECISION

# Bring It In-House

- Need deep signal-level understanding
- Fast iteration loop: test → visualize → fix → retest
- Full ownership of the data pipeline

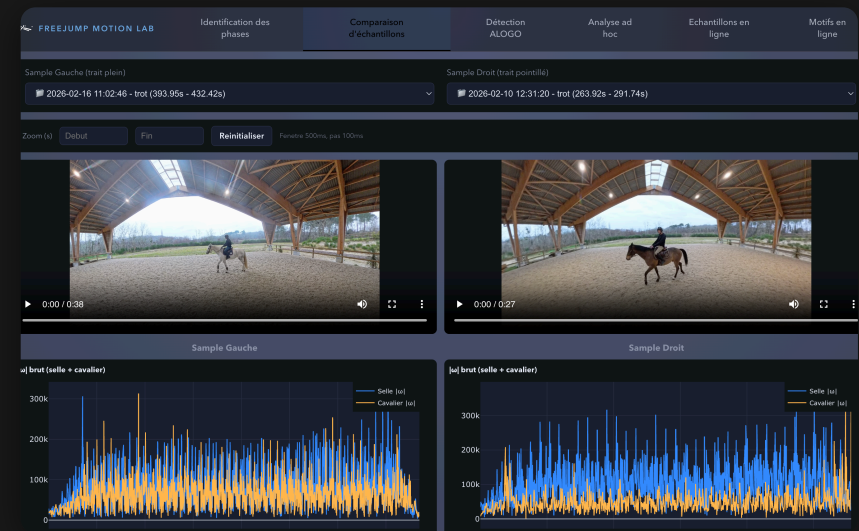
Outsourcing the algorithm means outsourcing understanding.  
We need to **see** the data to build a robust algorithm.

## APPENDIX — PROPRIETARY TOOL

# Freejump Motion Lab

Developed internally by the R&D team

- IMU visualization synchronized with video
- Multi-sensor comparison (saddle vs rider)
- Timeline annotations
- Pattern detection (Matrix Profile)
- Automatic phase analysis
- Cross-sensor correlation



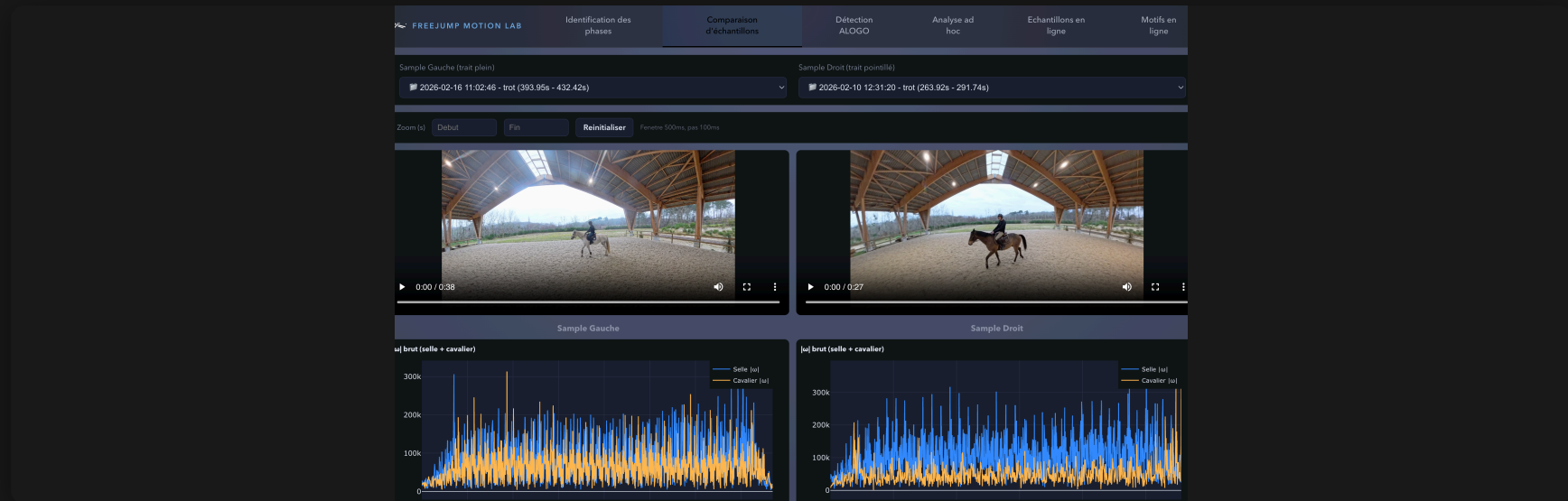
# What the Tool Enables

## Explore

- Fine analysis of inertial signals and their energy signatures alongside video
- Real-time saddle vs rider comparison
- Visual identification of key events

## Analyze

- Machine-learning-based learning workflows
- Automatic phase and transition detection
- Cross-sensor correlation and cross-validation



## APPENDIX — VISION

# Ready to Build

- **Understand the problem**

Alogo → key parameters identified

- **Build the right tools**

Freejump Motion Lab

- **Explore technologies**

IMU fusion · UWB · multi-sensor

- **Build the robust algorithm**

Validated features + iteration tools → GO

APPENDIX — PLANNING

# Product Roadmap

1

## Tubeless Vest

Tubeless Technology

✓ Completed

2

## Electronics Unit

IMU + sensor fusion

In Progress

Device integration  
Embedded algorithms

Target: end of year

3

## Gas Generator

Pyrotechnic cartridge

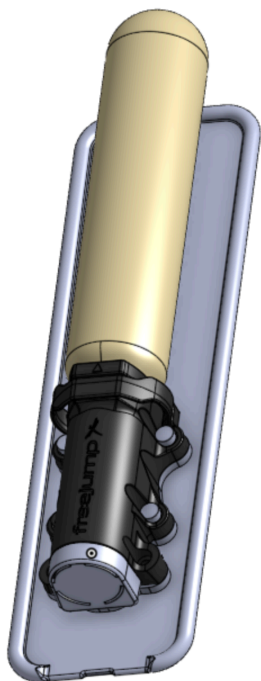
In Progress

Ineris declassification

Target: October 2026



## Gas Generator — Key Partnerships



### AQtech

Started from a wireless motorcycle airbag concept and adapted it to our system · Exterior design kept aligned with the existing product while integrating a pyrotechnic cartridge · Ogive-shaped piston needle update · Sealing and connector reliability work

### Astotec

Overmolding around the pyrotechnic charge · Based on AQtech's original pyro support geometry

### Groupfu

Machining of metal parts: cartridge body and piston

### Manuthiers

Injection molding of plastic parts: generator body, back plate, clip cap, cartridge clip · Overmolded cable passing through the back plate

